

Diploma in

PHOTOSHOP



SUMMARY NOTES

Scale & Perspective
Lessons 19 & 20

PHOTOSHOP CONTAINS SEVERAL TOOLS TO ADJUST SCALE AND PERSPECTIVE IN OUR IMAGES. IN THIS LESSON WILL BE LOOKING AT SOME OF PHOTOSHOP'S FEATURES SUCH AS CONTENT-AWARE SCALE TO RESIZE AN OBJECT AND MOVE IT TO ANOTHER AREA WITHIN THE IMAGE, AS WELL AS PUPPET-WARP TO MANIPULATE PERSPECTIVE.

2.01 Scale and Distort

2.02 Puppet Tool

2.03 Perspective Tool

2.04 Content Aware Family

Scale & Perspective

SCALE AND DISTORT

“To apply non-destructive transformations to raster images, use Smart Objects. Transforming a vector shape or path is always non-destructive.”

Transforming scales, rotates, skews, stretches, or warps an image. You can apply transformations to a selection, an entire layer, multiple layers, or a layer mask. You can also apply transformations to a path, a vector shape, a vector mask, a selection border, or an alpha channel. Transforming affects image quality when you manipulate the pixels.

To apply non-destructive transformations to raster images, use Smart Objects. Transforming a vector shape or path is always non-destructive because you're only changing the mathematical calculations producing the object.

To make a transformation, first select an item to transform and then choose a transformation command. If necessary, adjust the reference point before manipulating the transformation. You can perform several manipulations in succession before applying the cumulative transformation.

For example, you can choose Scale and drag a handle to scale, and then choose Distort and drag a handle to distort. Then press Enter or Return to apply both transformations.

To make a transformation, first select an item to transform and then choose the Transform Tool

- **Free Transform:** Ctrl/Cmd T
- **Edit > Free Transform**
- **Edit > Transform Option**

When using the Transform tool you will have specific options to use:

Scale: Enlarges or reduces an item relative to its reference point, the fixed point around which transformations are performed. You can scale horizontally, vertically or both

Rotate: Turns an item around a reference point.

Skew: Slants an item vertically and horizontally.

Distort: Stretches an item in all directions.

Perspective: Applies one-point perspective to an item.

Warp: Manipulates the shape of an item.

Rotate 180, Rotate 90 CW, Rotate 90 CCW: Rotates the item by the specified number of degrees, either clockwise or counter-clockwise.

Flip: Flips the item vertically or horizontally.

Note: To choose a new option while transforming right click

PUPPETWARP

Puppet Warp provides a visual mesh that lets you drastically distort specific image areas, while leaving other areas intact. Applications range from subtle image retouching (such as shaping hair) to total transformations (such as repositioning arms or legs).

To access Puppet Warp please go to Edit> Puppet Warp.

Options

ShowMesh: Deselect to show only adjustment pins, providing a clearer preview of your transformations.

Mode: Determines the overall elasticity of the mesh. Choose Distort for a highly elastic mesh good for warping wide-angle images or texture maps.

Density: Determines the spacing of mesh points. More Points increases precision but requires more processing time; Fewer Points does the opposite.

Expansion: Expands or contracts the outer edge of the mesh.

Hide pins: Press H to hide pins

Remove Pins: Ctrl/Cmd + Backspace, Ctrl/Cmd + Delete, Alt/Option and press on pin,

Rotate Pins: Alt/Option (just outside pins)

Move Pins: Select and Drag

Note: If we don't put enough pins in, we'll find that the parts of our image that we're not working on, will start to move about, so pin them down.



PERSPECTIVE WARP

Sometimes, an object may look different in an image from how it appears in real life. This mismatch is due to perspective distortion. Images of the same object captured from different camera distances and angles of view exhibit different perspective distortion. Photoshop lets you easily adjust perspective in images. This feature is particularly useful for images having straight lines and flat surfaces—for example, architectural images and images of buildings. You can also use this feature to composite objects having different perspectives in a single image.

To access Perspective Warp please go to Edit> Perspective Warp.

- Draw Grids needed in image while in Layout mode.
- To add multiple grids together, connect grids together in Layout Mode
- To manipulate perspective swap to Warp mode.
- Choose points to warp and move individually in Warp Mode.
- Once complete hit ok or enter button.

Options

- Automatically level near horizontal lines
- Automatically straighten near vertical lines
- Automatically straighten both vertically and horizontally



CONTENT AWARE FAMILY

When using the Content Aware Family you will have specific options to use:

- **Content Aware Scale**
- **Content Aware Fill**
- **Content Aware Crop**
- **Content Aware Move**

Content Aware Scale

Content-Aware Scale resizes an image without changing important visual content such as people, buildings, animals, and so forth. While normal scaling affects all pixels uniformly when resizing an image, content-aware scaling mostly affects pixels in areas that don't have important visual content Edit > Content-Aware Scale.

Note: To protect certain areas from scaling you can make a selection around the content you want to protect and then, in the Channels panel, click Save Selection As Channel.



Content-Aware Fill

The Content-Aware Fill in uses nearby content for seamless blending with the surrounding content.

1. Select the area you want to fill.
2. Use Edit>Fill or right click>Fill
3. Choose Content aware from drop down box.

Content Aware Crop

Content Aware Crop works like Content Aware Fill and Content Aware Scale by blending with the surrounding content.

To use you must have the option turned on with the Crop tool.

Crop outwards of the image.

Photoshop will try match and add the pixels for you.

Content-Aware Move

Use the Content-Aware Move tool to select and move a part of a picture. The image is recomposed, and the hole left behind is filled using matching elements from the picture.

Options

- **Move:** Used to place objects in different locations (most effectively when the background remains similar).
- **Extend:** Used to Copy images to another location. Can also be used to expand or contract objects such as hair, trees, or buildings.
- **Structure:** Enter a value between 1 and 7 to specify how closely the patch should reflect existing image patterns.
- **Color:** Enter a value between 0 and 10 to specify the extent to which you want Photoshop to apply algorithmic color-blending to the patch.
- **Sample All Layers:** Enable this option to use all layers to create the result of the move in the selected layer.
- **Transform On Drop:** When this option is enabled, you can scale the part of the image that you've just moved.





THANK YOU